

EXPERIMENT NO. :- 09

OBJECT :- Program And Algorithm In C For Binary Search.

Binary Search Program

```
#include<stdio.h>

#define size 20

int binary_search(int arr[] , int n , int target)
{
    int start = 0 , end = n -1 ;
    int mid ;
    while(start <= end)
    {
        mid = start + (end - start)/2;
        if(arr[mid]==target)
        {
            return mid; // element is found return index
        }
        else if(arr[mid] > target){
            end = mid - 1;
        }
        else{
            start = mid + 1;
        }
    }
    return -1; // target element is not found
}

int main()
{
    int arr[size] , n , target ;

    printf("\n enter the number of element in the array = ");
```

```

scanf("%d",&n);

printf("\n enter the data in the array : ");

for(int i = 0 ; i < n ; i++)
{
    printf("\n enter the arr[%d] element : ",i+1);
    scanf("%d",&arr[i]);
}

printf("\n enter the target element : ");
scanf("%d",&target); // target element

int pos = binary_search(arr , n , target);

if(pos==-1){
    printf("element is not found ");
}

else{
    printf("element is found at %d position",pos+1);
}
}

```

OUTPUT :

enter the number of element in the array = 5

enter the data in the array :

enter the arr[1] element : 1

enter the arr[2] element : 2

enter the arr[3] element : 3

enter the arr[4] element : 4

enter the arr[5] element : 5

enter the target element : 3

element is found at 3 position

ALGORITHM:

BINARY_SEARCH(A, lower_bound, upper_bound, TARGET)

Step 1: [INITIALIZE] SET START = lower_bound

END = upper_bound

Step 2: Repeat Steps 3 and 4 while START <= END

Step 3: SET MID = START + (END – START)/2

Step 4: IF A[MID] = TARGET

RETURN MID

Go to Step 6

ELSE IF A[MID] > VAL

SET END = MID - 1

ELSE

SET START = MID + 1

[END OF IF]

[END OF LOOP]

Step 5: RETURN -1

Step 6: EXIT